RESEARCH PROGRESS / MEETING COPY

When can we trust recovery?

LiDAR recovery reliability in GPS-denied navigation

CONTROLLED EVALUATION COMPLETE

We tested whether existing LiDAR health signals recognize the return of accurate movement estimation after difficult corridor geometry.

48
paired geometries

192
completed scene runs

2
odometry systems

The project in one minute

We began with a simulated safe-landing baseline, then narrowed the research to a specific reliability question. We built a controlled LiDAR/IMU evaluation, calibrated on development data, froze the settings, and completed experiments in FAST-LIO and Point-LIO.

Main finding: fixed health policies show different trade-offs between premature reassurance and missed recovery declarations in the evaluated geometries.

Where we stand: experiments, result accounting and failure analysis are complete. The physical explanation of some patterns, publication novelty and the full manuscript remain unfinished.

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Prepared 19 October 2026. Scientific evidence checkpoint: 12 October 2026.
Repository refreshed to 5971024, including the 19 October test additions.

From landing to localization

We narrowed a broad drone problem to a measurable question about estimation reliability.

Original implementation

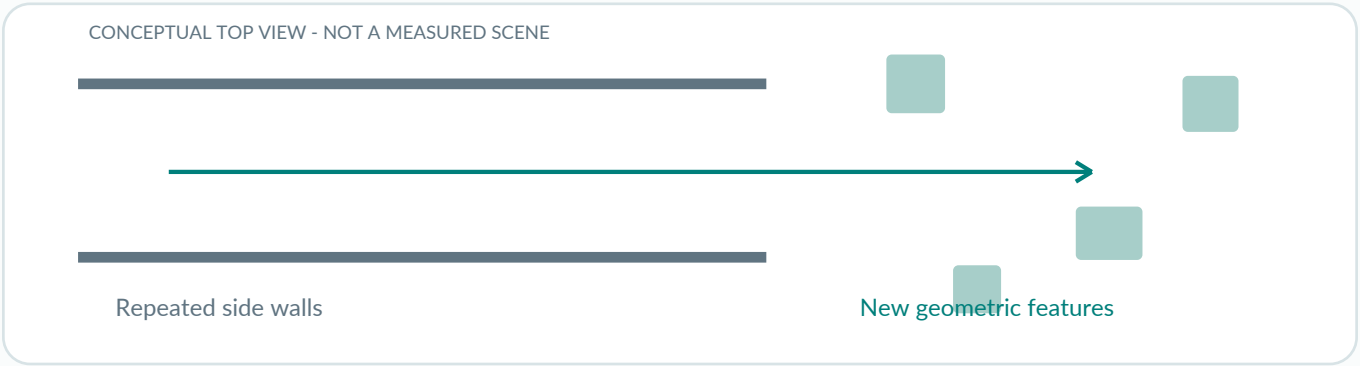
Synthetic terrain + noisy distance sensing + a height map. Check slope, roughness, obstacles and the full vehicle footprint before selecting a target.

Why the scope changed

Navigation, mapping, landing and control are separate problems. A focused study can isolate one scientific question using existing software and controlled data.

Archived baseline: 100 synthetic cases; all produced targets passed synthetic truth checks. Targets were available in 45/80 cases containing safe terrain (56.25%). These findings belong to the earlier landing study.

Why a corridor is difficult



Similar-looking surfaces give weak clues about movement along the corridor. This is **geometric degeneracy**. On exit, richer geometry may improve new movement estimates while earlier position drift remains.

Geometry

Are useful movement constraints present?

Local motion

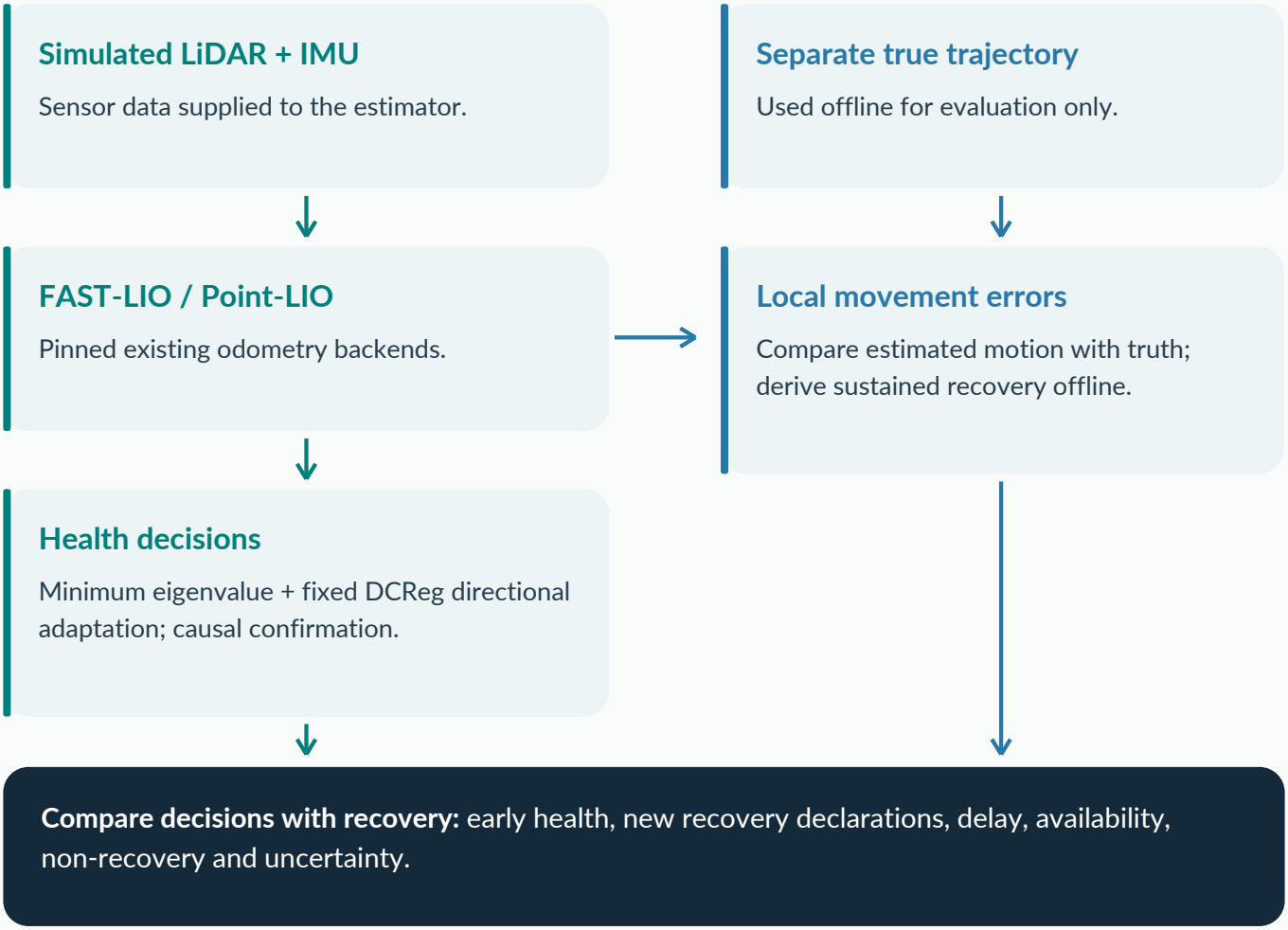
Are the newly estimated movements accurate?

Total drift

How much earlier position error remains?

A complete evidence workflow

Existing estimators, read-only diagnostics and a separate reference-based evaluator.



Why simulation is the primary evidence

GEODE: development replays did not supply usable recovery evidence; failures were retained.
Hilti-Oxford Exp18: a development replay worked, but reference coverage and independence were insufficient for sustained final recovery claims.
Simulation: supplies controlled geometry and known motion, with truth kept outside the estimator and online health decision.

We implemented diagnostic export, reference association, reset handling, resumable runners, input/output identities, failure accounting and figures. The DCReg detector adaptation excludes full DCReg registration mitigation.

03 / FAIR EXPERIMENTAL DESIGN

Freeze first. Evaluate second.

Development calibration and final-outcome evaluation were kept separate.

Short / narrow

12 pairs / corridor + matched feature control

Short / wide

12 pairs / corridor + matched feature control

Long / narrow

12 pairs / corridor + matched feature control

Long / wide

12 pairs / corridor + matched feature control

48 pairs x 2 scenes x 2 backends = 192 completed runs.
Each pair contains a corridor and a control with extra geometric features and matched motion. Both backends received identical corresponding sensor inputs. There are 48 shared geometry units, not 192 independent geometries.

Reference-defined recovery

Window 1
1 second

Window 2
1 second

Window 3
1 second

→ **Confirmed recovery**

Each consecutive, non-overlapping movement window must have translation error <= 0.20 m and rotation error <= 5 degrees. The three-window confirmation must finish within 20 seconds of the predefined exit. These are research tolerances.

Online health is separate: decisions use causal diagnostic data on a 10 Hz grid with three-tick confirmation. Offline recovery labels use reference motion; future reference samples never enter the online indicator.

Design disclosure: an earlier failed geometry screen informed the revised control spacing and candidate pools. Its IDs were excluded. Fresh scenes were selected using geometry criteria before estimator execution; final accuracy results did not select scenes or change thresholds.

Low early health can hide misses

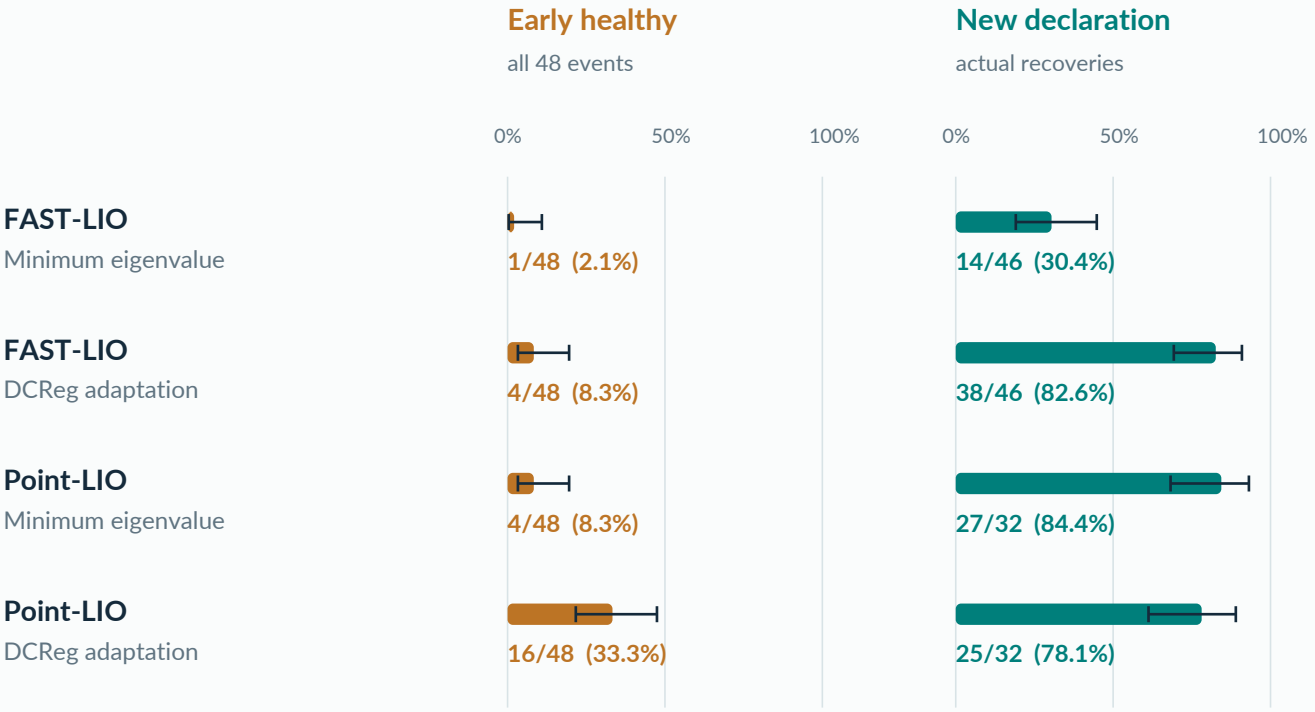
The locked policies show different trade-offs under the evaluated simulation design.

FAST-LIO: 46 / 48 recovered

Two did not recover within the horizon.

Point-LIO: 32 / 48 recovered

Sixteen did not recover within the horizon.



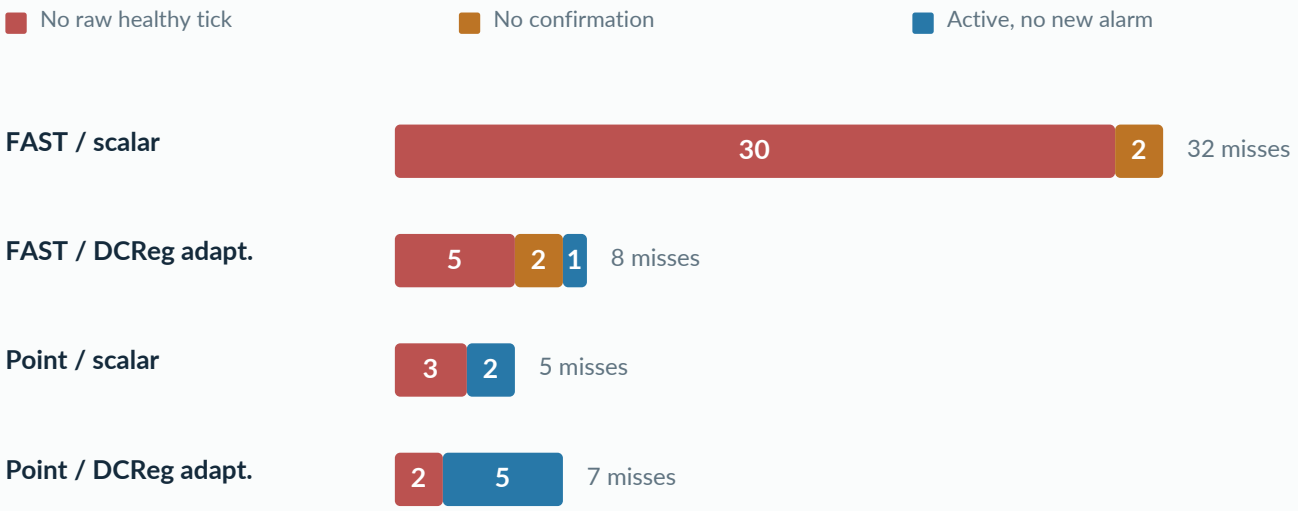
Early healthy means active health before sustained-recovery onset; it is not proof of instantaneous pose error. Bars show event rates; whiskers are 95% Wilson intervals. Early health uses **all 48 events**; detection uses each backend's own recovered events. These denominators differ and do not form an estimator leaderboard.

Example: FAST-LIO minimum eigenvalue had only 1/48 early-healthy events, but declared recovery in just 14/46 recovered cases. A cautious policy can miss useful recovery notifications.

Detected-only median delays: FAST scalar 0.85 s; FAST DCReg adaptation 6.80 s; Point scalar 3.40 s; Point DCReg adaptation 3.00 s. Misses have no observed delay, so a short median does not imply better overall responsiveness. Full horizon decision availability was 100% for all four policies.

A missed alarm has several meanings

Post-result analysis reproduced the original labels and kept all frozen counts unchanged.



Endpoint distinction: five of the seven Point-LIO DCReg-adaptation misses had an active healthy state during the recovered interval, but no new qualifying transition. A missed new declaration does not always mean continuous unhealthy status.

Recovery also varies across geometry groups

	Short / narrow	Short / wide	Long / narrow	Long / wide
FAST-LIO	12/12	10/12	12/12	12/12
Point-LIO	12/12	4/12	12/12	4/12

Point-LIO recovered in 24/24 narrow versus 8/24 wide corridors. Emphasizing width after seeing outcomes is **exploratory**; the data do not establish a causal width law. Each group has only 12 geometries.

Measurement-count summaries do not support a simple count-only explanation: FAST scalar's missed and detected groups had almost identical median accepted counts. Directional distribution, visibility, correspondence quality and estimator history remain possible explanations.

The scientific argument is unfinished

Implementation acceptance, scientific novelty and real-world reliability are separate questions.

COMPLETE	Controlled simulation, two backends and diagnostic recording
COMPLETE	Development calibration, protocol freeze and final 192 runs
COMPLETE	Scoped result-accounting audit, failure analysis and figures
OPEN	Physical explanation of geometry-dependent outcomes
OPEN	Precise contribution beyond the closest existing literature
OPEN	Full manuscript; main source still contains a proposal abstract

Difficulties that determine the next work

1

Distinct contribution

OCE already studies local-motion usability and recovery delay. Our candidate contribution concerns fixed-policy transfer across fresh geometry groups in two LIO backends. Its usefulness and novelty need scientific review.

2

Mechanism and endpoint interpretation

Width patterns are observations. We still need to explain the behavior and clarify when an application needs active health versus a new recovery notification.

3

Fair comparison and generalization

The directional method is a detector adaptation. Broader claims need a faithful comparator and suitable additional validation. Current real recordings do not establish sustained recovery.

Claim boundary: the evidence applies to the evaluated simulated straight-corridor design and fixed settings. It does not establish a new detector, estimator superiority, universal reliability or flight safety. The scoped audit is not journal peer review.

Turn the evidence into a paper

Proposed sequence for professor feedback; no new experiment is implied by this briefing.

1 Write the full draft

Present the question, frozen design, results, failures and limitations using the existing evidence.

2 Resolve contribution and endpoint

Compare closely with OCE/DCReg and explain active health versus a new declaration without replacing the frozen headline results.

3 Choose a justified validation step

Assess comparator feasibility. If one mechanism matters to the claim, freeze a focused study on fresh scenes before collecting outcomes.

4 Obtain scientific review and revise

Ask whether the empirical finding is distinct and useful, and which additional evidence would resolve a specific objection.

> **Questions for ma'am:** Is the transfer trade-off a sufficient contribution? Should we emphasize active health, new declarations or both? Would a closer comparator, focused simulation or better real reference add the most value?

The final cohort is now exposed. New explanations are exploratory; confirmation needs a prespecified hypothesis, controls, endpoints and fresh geometry IDs. No retuning on final outcomes.

Evidence index - clickable repository sources

S1 Original landing prototype

S2 Research goal and evidence boundaries

S3 Accepted final results and numerical exports

S4 Independent result-accounting audit

S5 Exploratory failure investigation

S6 Candidate contribution and literature overlap

S7 Claim-to-evidence map

S8 Current manuscript source

S9 Metrics and recovery definitions

Charts read accepted CSVs directly; no estimator was rerun. Early-health denominators on p.5 use all 48 events; the accepted report also provides conditional rates. The PDF sources manifest records the input hashes and revision.